

CONCLUSION

Final Product : Smart Sole Cleaner - Model BE

After evaluating the performance, cost-efficiency, and practicality of all proposed variants, we have finalized **Model BE** as our chosen design for further development and implementation.

Components:

- Fan
- Brush
- Heating Coil
- Dust Collector (without fan)

Key Features of Sole Dirt Removal – Model BE:

1. **Efficient and Cost-Effective:**
Uses a fan instead of a compressor, reducing system cost while maintaining good cleaning performance.
2. **Improved Cleaning with Brush:**
Brush mechanism removes mud and dirt more effectively from deep treads compared to a sponge.
3. **Low Power Consumption:**
Operates with reduced energy demand due to the use of a fan and a simple (non-powered) dust collector.
4. **Simple Dust Collection System:**
Dust collector without a fan keeps the design minimal and power-efficient while still effectively collecting debris.
5. **Portable Heating Coil:**
Dries moisture from soles quickly, reducing the risk of slippery floors and enhancing hygiene.
6. **Automatic Operation:**
Starts cleaning as soon as the user steps on the meshed area, providing a hands-free experience.